

SPE Class X-Ray Solar Proton Storm Shelter Protocol

1. Early Warning Triggers

Deep space heliospheric sensors trigger alarm code BRAVO upon detecting Class X flare coronal mass ejections with proton flux exceeding 1000 pfu at >10 MeV.

2. Crew Relocation Timeline

All extravehicular activities are aborted immediately. Crew must complete ingress into the sub-surface habitat central storm shelter within 45 minutes of primary alert.

3. Storm Shelter Shielding

The shelter is surrounded by 50 cm polyethylene water storage bladders and 1.2 meters of compacted Martian regolith, reducing cumulative dose rates to under 0.15 mSv/day.